

Observation Discriminant and Spherical Fold Image of the Orthogonal-Circle Ruled Surface

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Abstract

We determine the full critical locus of the planar observation map attached to the orthogonal-circle ruled surface. Its Jacobian factors as

$$J_F = -\frac{2 \sin t}{Q^3} (A(c) + uB(c)),$$

so the critical set is not only the previously isolated rational branch: it also contains the complete symmetry segment $t = 0$. The symmetry segment is an ordinary Whitney fold for $u > 0$. The two signed rational arms are ordinary folds away from their common initial point and occupy $c_b \leq c \leq 1$, where $c_b = 0.613055096856276\dots$ is the unique root in $(0, 1)$ of $c^5 - 5c^4 + 6c^3 - c^2 + c - 1$. At $(t, u) = (0, 0)$ the two critical branches meet and the ordinary-fold condition fails; locally $J_F = 2t(u - t^2) + \text{higher-order terms}$. Their critical values are different objects from the normalized surface-position curve: the symmetry fold maps to the segment $N = 0$, while the rational arms share one discriminant curve tangent to that segment at $(N, M) = (0, 1)$.

For completeness we retain the parameter-free spherical radial image $\Phi = S/\|S\|$ of the rational branch. It consists of two mirrored arcs that start regularly at $(1, 0, 0)$ and meet the parameter boundary at

$$\left(\sqrt{\frac{2 - c_b}{2}}, 0, \sqrt{\frac{c_b}{2}} \right) \approx (0.8327499334, 0, 0.5536493009).$$

This radial image is not identified with the observation discriminant or the Gauss map. Its third spherical component is monotone and maximal at the boundary. Exact polynomial identities, Sturm counts and numerical checks are reproduced by a standard-library verifier. No Jones-spinor, quantum, spinorial or Atiyah–Patodi–Singer interpretation is claimed.

1 Correction scope and claim status

Let $c = \cos t$, $s = \sin t$ and $Q = \sqrt{c(2 - c)}$, with $t \in [-\pi/2, \pi/2]$ and $u \in [0, 1]$. The ruled surface of [1] is

$$S(t, u) = \begin{pmatrix} c + 2u(1 - c) \\ (1 - u)s \\ uQ \end{pmatrix}. \tag{1}$$

Four distinct objects must be kept separate:

1. the component-zero curve $\mathcal{L}_N = \{N = 0\}$, given away from $c = 1$ by $u = (1 - c)/2$;
2. the isolated immersion zeros, obtained only after all components of $S_t \times S_u$ vanish; and

3. the full critical locus $\text{Crit}(F) = \{J_F = 0\}$ of the planar observation map;
4. the discriminant $F(\text{Crit}(F)) \subset \mathbb{R}^2$ and the separate radial sphere map $S/\|S\| \subset S^2$.

Only the image of the third object under F is the observation discriminant. Radial normalization of the surface position is a different, origin-dependent construction.

Statement	Status here	Input
Full critical-locus decomposition	theorem	exact factorization of J_F
Fold status away from $(0, 0)$	theorem	kernel transversality/Sturm count
Degenerate intersection at $(0, 0)$	theorem	local expansion
Observation discriminant	theorem	restriction of F
Spherical radial fold image	theorem	(1)
Jones, spinor or APS interpretation	not claimed	absent from this geometry

2 Observation map and full critical locus

The complex observation field used in [1] is $F = N + iM$, where

$$N(c, u) = (1 - c)(1 - c - 2u), \quad (2)$$

$$M(c, u) = \frac{(1 - u)c^2(2 - c) - u(1 - c)^2(1 + c)}{Q}. \quad (3)$$

Its fold Jacobian is $J_F = N_t M_u - N_u M_t$.

Theorem 2.1 (Full critical locus). *Where $Q > 0$, the Jacobian factors exactly as*

$$J_F = -\frac{2s}{Q^3}(A(c) + uB(c)), \quad (4)$$

where

$$A(c) = (1 - c)c(2 - c)(1 + 2c - c^2), \quad B(c) = c^3 - 3c + 1.$$

Consequently the critical locus in the parameter strip has two components:

$$\mathcal{C}_0 = \{(t, u) : t = 0, 0 \leq u \leq 1\}, \quad (5)$$

$$\mathcal{C}_1 = \{(t, u) : A(\cos t) + uB(\cos t) = 0\}. \quad (6)$$

Proof. Direct differentiation and collection in powers of u gives (4). In the real parameter interval, $s = 0$ is exactly $t = 0$. The other factor gives $\mathcal{C}_1$. Neither factor may be cancelled when determining the full zero set.

Proposition 2.2 (Rational critical branch). *Away from $s = 0$ and the vertical asymptote $B(c) = 0$, the component $\mathcal{C}_1$ is*

$$u = u_{\text{fold}}(c) := -\frac{(1 - c)c(2 - c)(1 + 2c - c^2)}{c^3 - 3c + 1}. \quad (7)$$

The curve $u = (1 - c)/2$ is instead $\mathcal{L}_N = \{N = 0\}$ and is not equal to (7).

Proof. Solving $A + uB = 0$ proves (7). The distinction is exact: at $c = 4/5$,

$$\frac{1 - c}{2} = \frac{1}{10}, \quad u_{\text{fold}}(c) = \frac{392}{925}.$$

Write

$$n(c) := -(1-c)c(2-c)(1+2c-c^2), \quad (8)$$

$$d(c) := c^3 - 3c + 1, \quad (9)$$

so $u_{\text{fold}} = n/d$.

Proposition 2.3 (Physical fold interval). *There is a unique $c_b \in (0, 1)$ satisfying*

$$p_b(c_b) = 0, \quad p_b(c) := c^5 - 5c^4 + 6c^3 - c^2 + c - 1. \quad (10)$$

Numerically,

$$c_b = 0.613055096856276 \dots$$

The part of (7) lying in $0 \leq u \leq 1$ is exactly $c_b \leq c \leq 1$, with $u(c_b) = 1$ and $u(1) = 0$.

Proof. The equation $u_{\text{fold}} = 1$ is $n - d = 0$, which expands to (10). The Sturm variations of p_b at 0 and 1 are respectively 4 and 3, hence there is exactly one root in $(0, 1)$. It is isolated by $613/1000 < c_b < 614/1000$. On $[c_b, 1]$ the denominator has no zero, $d < 0$, and the inequalities $0 \leq n/d \leq 1$ follow from the endpoint signs and uniqueness of the root of $n - d$.

Theorem 2.4 (Ordinary folds and the exceptional intersection). *Using the standard Whitney fold convention and its plane-to-plane criterion [2, 3], the critical points have the following types.*

1. Every point of $\mathcal{C}_0$ with $u > 0$ is an ordinary Whitney fold of F .
2. Every point of the physical part of $\mathcal{C}_1$ with $c_b \leq c < 1$ is an ordinary Whitney fold, using a smooth extension across $u = 1$ at the parameter boundary.
3. At $(t, u) = (0, 0)$ the two critical components meet transversely and the ordinary-fold condition fails.

Proof. On $\mathcal{C}_0$,

$$F(0, u) = (0, 1 - u), \quad F_u(0, u) = (0, -1).$$

The kernel of DF is the t direction, and direct differentiation gives $N_{tt}(0, u) = -2u$. Hence

$$\det(F_u, F_{tt}) = -2u \neq 0$$

for $u > 0$, which is the fold criterion.

On $\mathcal{C}_1$, let $v = (N_u, -N_t)$ span $\ker DF$. Since $J_F = (-2s/Q^3)(A + uB)$, restriction to $A + uB = 0$ gives

$$dJ_F(v) = -\frac{4s^2}{Q^3 B(c)} C_8(c), \quad (11)$$

where

$$C_8(c) = (c - 1)R_7(c)$$

and

$$R_7(c) = 3c^7 - 8c^6 - 12c^5 + 60c^4 - 69c^3 + 30c^2 - 6c - 1.$$

The Sturm variation of R_7 is the same at $613/1000$ and 1, so it has no root there; $R_7(1) = -3$. Also B has no root on this interval. Thus (11) is nonzero for $c_b \leq c < 1$.

Finally set $\delta = 1 - c$. Exact substitution gives

$$A = 2\delta - 3\delta^3 + \delta^5, \quad B = -1 + 3\delta^2 - \delta^3.$$

Since $\delta = t^2/2 + O(t^4)$,

$$J_F(t, u) = 2t(u - t^2) + O(|t|^5 + |u||t|^3). \quad (12)$$

The two local critical branches are therefore $t = 0$ and $u = t^2 + O(t^4)$, with distinct tangent lines. At their intersection DF has rank one but $\det(F_u, F_{tt}) = 0$, so the point is not an ordinary fold. No finer singularity label is imposed here.

Definition 2.5 (Observation discriminant). The observation discriminant is

$$\Delta_F := F(\text{Crit}(F)) \subset \mathbb{R}^2.$$

Proposition 2.6 (Two critical components, two critical-value branches). *The symmetry component maps exactly to*

$$F(\mathcal{C}_0) = \{(N, M) = (0, m) : 0 \leq m \leq 1\}.$$

The two signed arms of $\mathcal{C}_1$ have the same image because F is even in t . Writing $\delta = 1 - c$ along that image,

$$u_{\text{fold}} = 2\delta + O(\delta^3), \quad (13)$$

$$N = -3\delta^2 + O(\delta^4), \quad (14)$$

$$M = 1 - 3\delta + \frac{3}{2}\delta^2 + O(\delta^3). \quad (15)$$

Consequently the rational discriminant is tangent to the symmetry segment at $(N, M) = (0, 1)$:

$$N = -\frac{1}{3}(1 - M)^2 + O((1 - M)^3). \quad (16)$$

Its other physical endpoint is

$$F(c_b, 1) = \left(c_b^2 - 1, -\frac{(1 - c_b)^2(1 + c_b)}{\sqrt{c_b(2 - c_b)}} \right) \approx (-0.6241634482, -0.2619196638).$$

Proof. At $t = 0$, substitution in N and M gives $(0, 1 - u)$. Evenness follows because N and M depend on t only through $c = \cos t$ and Q . The displayed expansions follow from the exact δ expressions in the proof of [theorem 2.4](#). Eliminating δ gives (16); setting $u = 1$ gives the final endpoint.

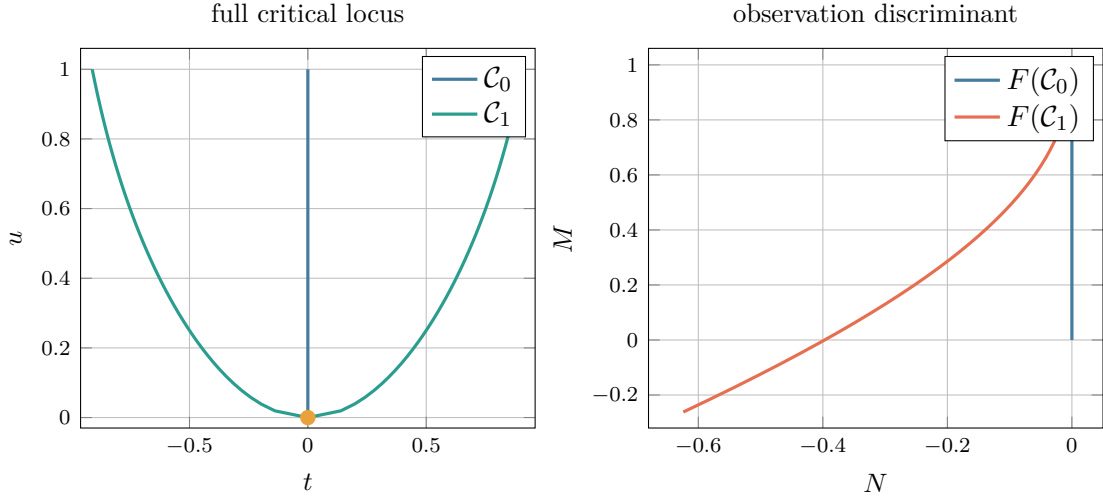


Figure 1: Left: the full critical locus contains the symmetry fold $\mathcal{C}_0$ and the two signed rational arms $\mathcal{C}_1$; they meet at the exceptional point (amber). Right: the symmetry fold maps to a line segment, while both rational arms share one critical-value curve tangent to that segment at $(0, 1)$.

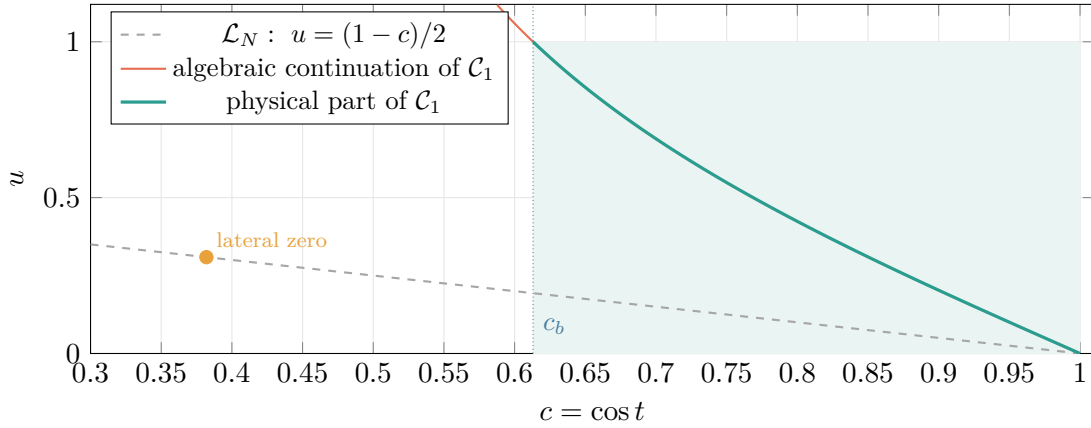


Figure 2: The component-zero curve $\mathcal{L}_N$ and the rational critical component $\mathcal{C}_1$ are different. The thick teal segment is the complete portion of $\mathcal{C}_1$ inside the parameter strip. The symmetry component $\mathcal{C}_0$, visible in (t, u) coordinates in figure 1, collapses to the boundary line $c = 1$ in this plot.

3 Spherical radial image of the rational component

Definition 3.1 (Spherical radial fold map). For $c \in [c_b, 1]$, put $u = u_{\text{fold}}(c)$ and define

$$X(c) = c + 2u(1 - c), \quad (17)$$

$$Y_{\pm}(c) = \pm(1 - u)\sqrt{1 - c^2}, \quad (18)$$

$$Z(c) = u\sqrt{c(2 - c)}, \quad (19)$$

$$R(c) = \sqrt{X(c)^2 + Y_{\pm}(c)^2 + Z(c)^2}. \quad (20)$$

The two normalized branches are

$$\Phi_{\pm}(c) := \frac{1}{R(c)}(X(c), Y_{\pm}(c), Z(c)) \in S^2. \quad (21)$$

The sign distinguishes $t > 0$ and $t < 0$; R is independent of that sign. We use the conventional coordinate labels (S_1, S_2, S_3) when plotting this unit vector on the Stokes–Poincaré sphere [5, 6], but no Jones-vector identity is assumed.

Remark 3.2 (Three maps, no automatic bridge). The observation discriminant $\Delta_F \subset \mathbb{R}^2$, the radial map $\Phi = S/\|S\|$, and the Gauss map

$$\nu = \frac{S_t \times S_u}{\|S_t \times S_u\|}$$

are distinct constructions. In particular, translating the surface changes Φ but leaves F , Δ_F and ν unchanged. Therefore neither a discriminant interpretation nor a polarization interpretation of Φ follows from normalization alone.

Theorem 3.3 (Endpoints). *The two branches have common endpoints*

$$\Phi_{\pm}(1) = (1, 0, 0), \quad (22)$$

$$\Phi_{\pm}(c_b) = \left(\sqrt{\frac{2 - c_b}{2}}, 0, \sqrt{\frac{c_b}{2}} \right) \quad (23)$$

$$\approx (0.832749933396492, 0, 0.553649300937099). \quad (24)$$

Proof. At $c = 1$, Proposition 2.3 gives $u = 0$, and (1) equals $(1, 0, 0)$. At $c = c_b$, $u = 1$, so $S = (2 - c_b, 0, \sqrt{c_b(2 - c_b)})$. Its squared norm is $2(2 - c_b)$, which gives the displayed unit vector.

Theorem 3.4 (Regular radial starting point). *The spherical radial fold map is regular at $t = 0$. With $c = \cos t$ and the sign of Y inherited from t ,*

$$\Phi(t) = \left(1 - \frac{t^2}{2} + O(t^4), t + O(t^3), t^2 + O(t^4) \right). \quad (25)$$

Consequently

$$\Phi'(0) = (0, 1, 0) \neq 0.$$

Proof. Expansion of (7) at $c = \cos t$ gives $u_{\text{fold}}(\cos t) = t^2 + O(t^4)$. Substitution in (1), followed by normalization, gives (25).

Remark 3.5. The former argument tested whether a one-dimensional curve surjects onto the two-dimensional tangent plane of S^2 . That is not the immersion criterion. Rank one is maximal for a curve, and theorem 3.4 supplies the required nonzero derivative.

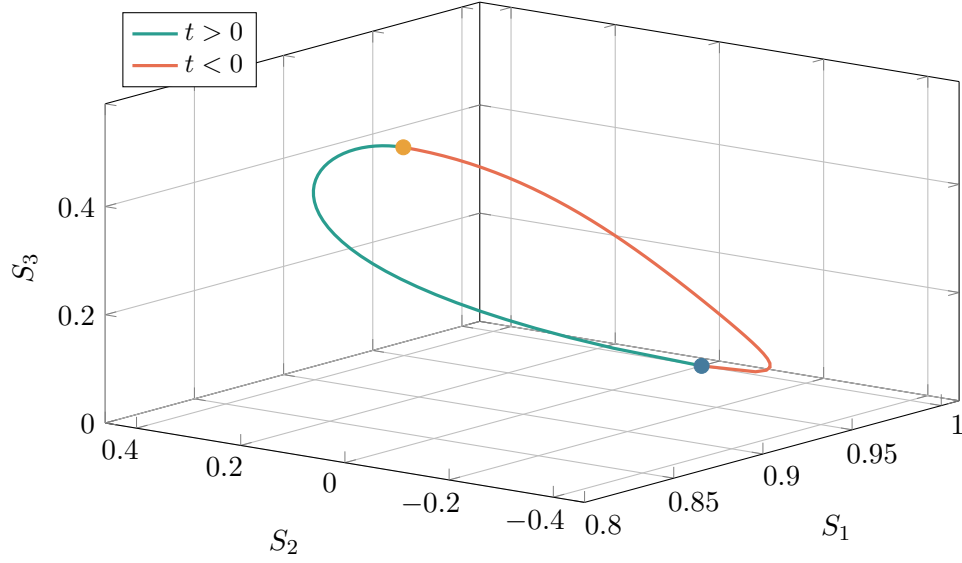


Figure 3: Spherical radial image of the physical part of $\mathcal{C}_1$. The two branches are mirror images across $S_2 = 0$ and share both the regular radial starting point (blue) and the parameter-boundary endpoint (amber). This is not the planar discriminant shown in [figure 1](#).

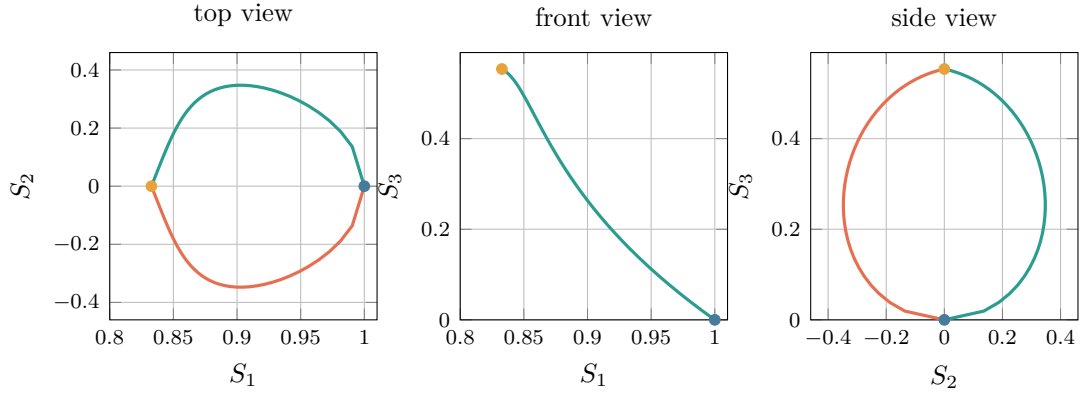


Figure 4: Three orthogonal views of the spherical radial image. These projections make the mirror symmetry and common endpoints visible without relying on a particular three-dimensional camera angle.

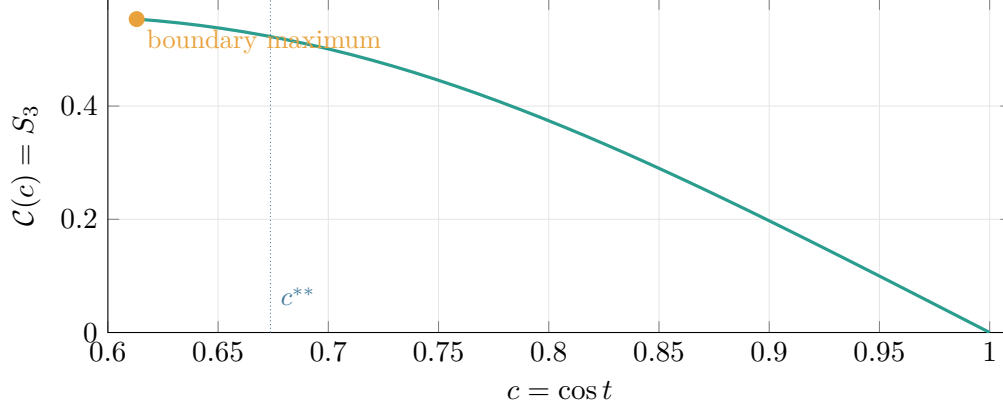


Figure 5: The third component of the spherical radial image. The former critical value $c_{\text{chiral}} \approx 0.3103$ lies outside the physical interval and is therefore absent. The dotted line marks the rational-critical-branch-skeleton intersection $c^{**} \approx 0.673741$. A circular-polarization interpretation is not assumed.

4 Boundary endpoint and third spherical component

Let $\mathcal{C}(c) := S_3(\Phi_+(c)) = Z(c)/R(c)$.

Theorem 4.1 (Boundary maximum). *On the physical fold interval $[c_b, 1]$, $\mathcal{C}(c)$ decreases strictly with c . Equivalently it increases from the regular start toward the parameter boundary. Hence*

$$\max_{c_b \leq c \leq 1} \mathcal{C}(c) = \mathcal{C}(c_b) = \sqrt{\frac{c_b}{2}} \approx 0.553649300937099. \quad (26)$$

There is no interior critical point of the third spherical component on the physical part of $\mathcal{C}_1$.

Proof. Writing $\mathcal{C}^2 = A/R_{\text{num}}$ with exact integer polynomials, direct differentiation gives

$$\frac{d}{dc} \mathcal{C}(c)^2 = \frac{n(c)Q_{17}(c)}{R_{\text{num}}(c)^2}. \quad (27)$$

The polynomial Q_{17} is given in [section A](#). Its Sturm variation is 9 at both $613/1000$ and 1 , so it has no root on that interval; $Q_{17}(4/5) > 0$. Since $n(c) < 0$ for $c_b < c < 1$, the derivative is negative. The endpoint value is [Theorem 3.3](#).

Remark 4.2 (What this boundary is not). The point $u = 1$ is a boundary point of the two-dimensional ruled-surface parameter strip. The construction supplies neither a spinor bundle on S^3 , a boundary Dirac operator, nor a self-adjoint boundary domain. Consequently it cannot by itself define an APS spectral projector. Any such identification would be an additional bridge.

5 Disposition of earlier claims

Earlier statement	Status in v5
$u_* = (1 - c)/2$ is the Whitney fold	False. It is the component-zero curve $\mathcal{L}_N$.
The rational branch is the full locus $\{J_F = 0\}$	False. The symmetry component $\mathcal{C}_0$ was omitted.
$\Phi = S/\ S\ $ is the observation discriminant	False without a bridge; the discriminant is $F(\text{Crit}(F))$.
The normalized curve is singular at $t = 0$	False. Its derivative is $(0, 1, 0)$.
Endpoint $(2/\sqrt{5}, \pm 1/\sqrt{5}, 0)$	Belongs to the auxiliary $\mathcal{L}_N$ curve, not the true fold.
Irreducible chirality quintic	Describes an interior maximum of that auxiliary curve; it is outside the physical true-fold interval.
$\sqrt{5}$ endpoint thread	An identity for the auxiliary curve only; removed from the radial-fold theorem.

The auxiliary $\mathcal{L}_N$ construction can still be studied under its correct name, but none of its constants may be transferred to the observation discriminant or radial fold image without an additional theorem.

6 Conclusion

The observation map has a two-component critical locus. The symmetry segment $\mathcal{C}_0$ consists of ordinary folds for $u > 0$; the physical rational component $\mathcal{C}_1$ consists of ordinary folds away from its common initial point. At $(0, 0)$ the components meet, the Jacobian has the local factor $2t(u - t^2)$, and ordinary-fold regularity fails. The discriminant is the union of a line segment and a rational critical-value curve with quadratic tangency at $(0, 1)$.

The separately defined spherical radial image retains the exact physical interval, endpoints, nonzero starting tangent and monotone third component proved in v4. It is origin-dependent and is not identified with the observation discriminant or the Gauss map. Thus no Jones, spinorial, quantum or APS conclusion follows from the present construction. A future Hopf lift must first choose and justify which sphere map is being lifted.

There remains one local singularity-theory problem. Put $x = 1 - M$ and $y = t$. Since $x_u(0, 0) = 1$, (x, y) are valid local source coordinates. With weights $\text{wt}(y) = 1$ and $\text{wt}(u) = 2$, direct expansion gives

$$x = u + \frac{1}{2}y^2 + O_w(4), \quad N = -uy^2 + \frac{1}{4}y^4 + O_w(6),$$

and hence, after eliminating u ,

$$(x, N) = \left(x, -xy^2 + \frac{3}{4}y^4 + O_w(6) \right).$$

Its displayed four-jet is equivalent by real rescalings to the specified jet $(x, xy^2 + y^4)$ occurring in the plane-to-plane recognition problem [4]. This does not classify the full germ: the exact map is even under $t \mapsto -t$, and finite determinacy and the higher-jet recognition conditions remain to be checked. The next well-posed problem is therefore an $\mathcal{A}$ -classification, or a symmetry-preserving $\mathbb{Z}_2$ classification, followed only then by a versal symmetry-breaking unfolding.

A Exact verification certificate

The companion file `verify_stokes_caustic_v5.py` uses only the Python standard library. It checks the exact Jacobian factors, the local $\delta = 1 - c$ expansions, the weighted map-germ jet, Sturm root counts, fold transversality, discriminant tangency, the physical interval, unit-sphere normalization, endpoint values, the nonzero radial starting derivative and third-component monotonicity.

For reproducibility, the derivative factor in [theorem 4.1](#) is

$$Q_{17}(c) = \sum_{j=0}^{17} q_j c^j,$$

with coefficients in increasing order

$$(q_0, \dots, q_{17}) = (0, -12, 14, 64, -346, 856, -810, 798, -4868, \\ 14160, -21630, 20432, -12800, 5462, -1588, 306, -36, 2).$$

The exact Sturm counts used in the paper are

polynomial	$V(\text{left})$	$V(\text{right})$	roots in interval
$p_b, (0, 1)$	4	3	1
$R_7, (613/1000, 1)$	3	3	0
$Q_{17}, (613/1000, 1)$	9	9	0.

References

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